In the syntax of First-Order Logic (FOL), which symbols are used to define an atomic sentence of the form P(t1, ..., tn)?

- A. Connectives and Constants
- B. Functions and Equality
- C. Equality and Quantifiers
- D. Variables and Quantifiers
- E. Predicates and Terms

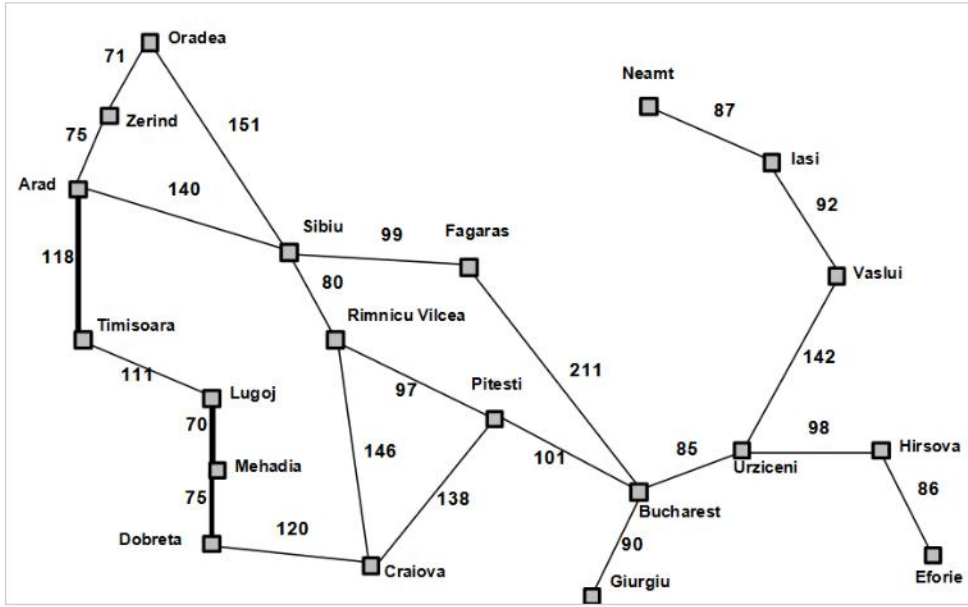
TRUE/FALSE 1 / 1

The agent function maps the agent's percepts (P) directly to actions (A), disregarding the history of percepts.

- True
- False

3 **CALCULATED NUMERIC** 0 / 6

Consider the problem of finding the shortest path from Mehadia to Bucharest using the A* algorithm with the straight-line distance heuristic. The graph consists of nodes representing cities and edges representing the distances between them.



Straight–line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Chapter 4, Sections 1–2

The evaluation function for A* is defined as f(x) = g(x) + h(x), where g(x) is the cost of the path from the start node to node x, and h(x) is the estimated cost of the cheapest path from node x to the goal node.
What is the value of f(Pitesti)?

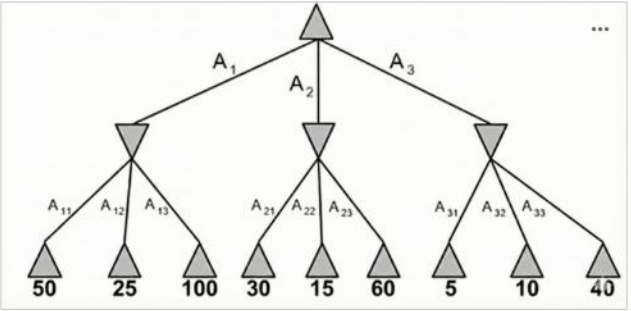
Your answer: 714

Variational Auto-encoders (VAE) are generative models composed of an encoder that maps input to a latent space and a decoder that generates the output or new data.

- [x] True
- [ ] False

5 **MULTIPLE CHOICE** **6 / 6**

Which branches in this game tree will be pruned by the Alpha-Beta algorithm.

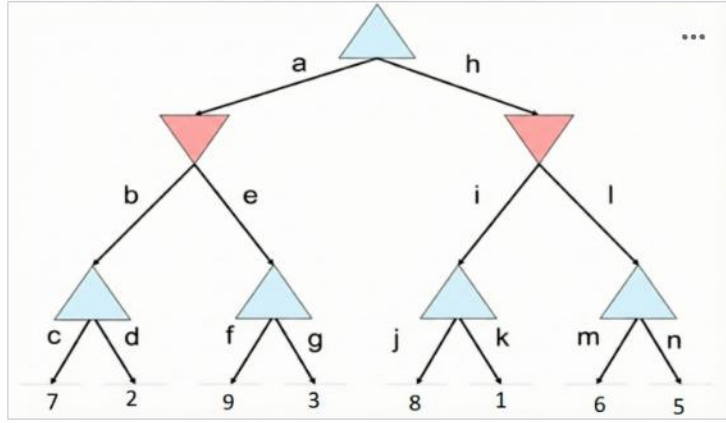


- [ ] **A.** A22, A23, A32, A33
- [ ] **B.** A22, A23
- [ ] **C.** A2, A3
- [ ] **D.** A12, A13, A23
- [x] **F.** A23, A32, A33

6 **CALCULATED NUMERIC**

5 / 5

In the following game tree, assuming the top node is a maximizing node and the red triangles are minimizing nodes, what will be the value in the root node?



Your answer: 7

7 **CALCULATED NUMERIC** 0 / 5

You are evaluating a classifier on 100 samples. 95 samples belong to Class A (Healthy) and 5 belong to Class B (Diseased).
If the classifier predicts all 100 samples as Class A, what is the resulting accuracy?
(State your answer as a decimal to two decimal places.)

Your answer: 95

8 **MULTIPLE CHOICE** 3 / 3

Which ethical concern is specifically highlighted regarding the societal implications of agent decisions?

- A. Lack of computational power
- B. Transparency and explainability of agent decisions
- C. Poor performance measures
- D. Limited scope for single-agent systems
- E. Inability to handle large datasets

9 **TRUE/FALSE** 2 / 2

The Rectified Linear Unit (ReLU) is a non-linear activation function that replaces all negative values in the convolution output with zero.

- (•) True
- ( ) False

10 **MULTIPLE CHOICE** 0 / 4

When generating a response during inference, such as a translation, a Transformer-based LLM produces the full output sequence by following which iterative process?

- ( ) **A.** Randomly selecting words until a grammatically correct sentence is formed.
- ( ) **B.** Selecting the entire sequence based on the highest accumulated probability score
- ( ) **C.** Outputting one word at a time, feeding that word back into the system to determine the next word until an end-of-sequence token is reached
- ( ) **D.** Using only the encoder output to calculate the full sequence of target probabilities.
- (•) **E.** Generating all words simultaneously in the output layer.

11 **TRUE/FALSE** 0 / 2

The Existential Quantifier (∃) is roughly equivalent to the conjunction (∧) of instantiations of a sentence.

- (•) True
- ( ) False

12 **TRUE/FALSE** 1 / 1

The Validation set is used to fine-tune and validate the model, while the Test set is used for the final evaluation of the model's performance.

- (•) True
- ( ) False

13 **TRUE/FALSE** 1 / 1

Artificial Neural Networks (ANNs) are primarily useful because they can model nonlinear and complex relationships between input and output data.

- (•) True
- ( ) False

14 **MULTIPLE CHOICE** 0 / 3

Which subfield of AI is associated with techniques such as Bayesian networks and Markov models?

- ( ) **A.** Uncertainty and Probabilistic Models
- ( ) **B.** Robotics and Perception
- ( ) **C.** Problem Solving and Search
- ( ) **D.** Knowledge Representation and Reasoning
- (•) **E.** Reinforcement Learning

15 **TRUE/FALSE** 2 / 2

Uninformed search strategies include Breadth-First Search, Uniform-Cost Search, and A* search.

- True
- False

16 **MULTIPLE CHOICE** 3 / 3

The primary challenge that makes an exact solution to a game like Chess infeasible for the Minimax algorithm is its reliance on which property?

- **A.** Exponential number of game states and depth of the tree
- **B.** Lack of a terminal test
- **C.** Optimality
- **D.** Time limits
- **E.** Requirement for an evaluation function

The Minimax algorithm is designed to achieve "perfect play" for which specific category of games?

- **A.** Deterministic, perfect-information games
- **B.** Cooperative games
- **C.** Nondeterministic games like Backgammon
- **D.** Games involving imperfect information like Poker
- **E.** Games involving three or more players

18 **TRUE/FALSE** 1 / 1

The key difference between game problems and standard search problems is that in games, the solution must be a strategy specifying a move for every possible opponent reply.

- True
- False

19 **MULTIPLE CHOICE** 3 / 3

In the linear equation y = m x + b + e what does the variable "y" represent in the context of linear regression?

A. The slope

B. The initial random value

C. The independent variable (input)

D. The error term (residual)

E. The dependent variable (output)

20 **MULTIPLE CHOICE** 3 / 3

In the K-means algorithm, after assigning all data samples to the nearest centroid, what is the next step in the iterative process?

A. Calculate the Gini index

B. Recalculate the centroids by calculating the mean of all data points in the cluster

C. Randomly select new initial centroids

D. Stop the iteration and output the final model

E. Generate a confusion matrix

What constitutes a valid "Stopping criteria" for generating a Decision Tree model?

- A. When the training data and test data are merged
- B. When the model achieves 100% precision
- C. When the number of leaves equals the number of features
- D. When the Gini index reaches 0.5
- E. When all data belongs to the same class

MULTIPLE CHOICE 3 / 3

The Gini index is an Attribute Selection Measure used in Decision Trees to quantify what?

- A. The likelihood of misclassifying a randomly chosen sample (disorder/impurity)
- B. The speed of model convergence
- C. The complexity of the dataset (

RUE/FALSE

0 / 1

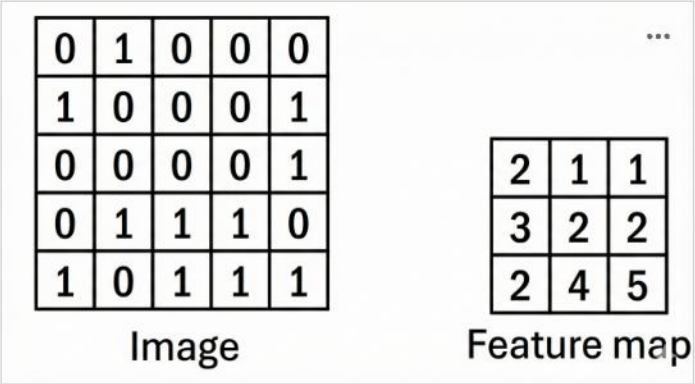
dversarial search is a technique commonly applied to pathfinding and optimization problems.

- True
- False

MULTIPLE CHOICE

6 / 6

Which of the filters, when applied to the given image using the convolution operation, will result in the provided feature map?
Assume a stride of 1 and no padding.



A.
1 1 1
1 0 1
1 0 1

B.
1 1 1
1 1 0
0 1 1

25 **MULTIPLE CHOICE** 3 / 3

How does Iterative Deepening Search (IDS) proceed to find a solution?

- A. By switching between Breadth-First and Depth-First search.
- B. By strictly following the lowest path cost g(n).
- C. By maintaining a closed set to avoid repeated states.
- D. By expanding the node closest to the goal based on h(n).
- E. By executing Depth-Limited Search iteratively starting from depth 0 to ∞.

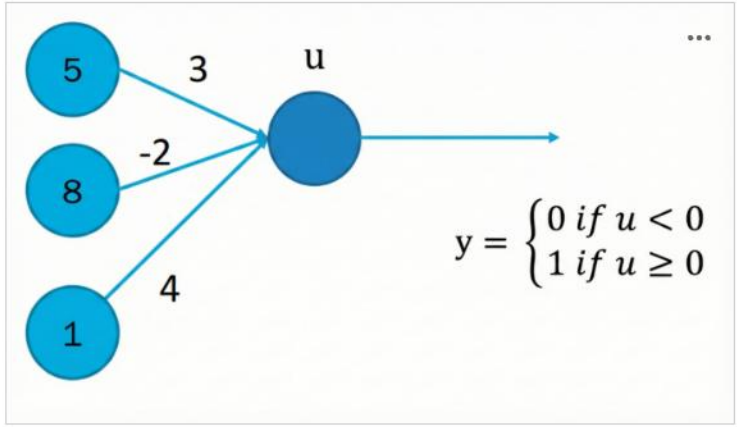
26 **CALCULATED NUMERIC** 5 / 5

A convolutional layer uses 6 distinct filters, each having a size of 2×2. If we consider only the weights (and ignore bias for simplicity), how many total weights are used in this layer?

Your answer: 24

27 **CALCULATED NUMERIC** 5 / 5

What is the output y of the following perceptron (enter a value):



Your answer: 1

MULTIPLE CHOICE 0 / 7

What is the CNF (Conjunctive Normal Form) of the expression $T \vee \neg((P \wedge Q) \Rightarrow R)$?
You may use the following logical equivalences for your conversion:

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

A. $(T \vee P) \vee (Q \wedge \neg R)$

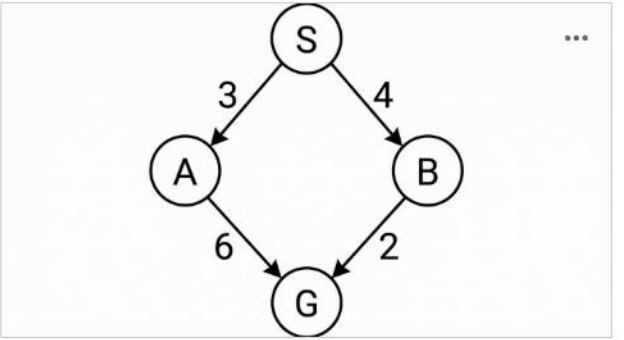
B. $(T \vee P) \wedge (T \vee (Q \wedge \neg R))$

C. $(T \vee P \vee Q) \wedge (T \vee \neg R)$

D. $(T \vee P) \wedge (T \vee Q) \wedge (T \vee \neg R)$

MULTIPLE CHOICE

Consider the following diagram where S is the start node and G is the goal node. The numbers on the edges represent the actual distance between nodes.



You are tasked to create an admissible heuristic for the A* algorithm. If h(S), h(A), and h(B) are the estimated costs to reach the goal G from S, A, and B respectively, which of the following sets of h-values is admissible?

A. h(S)=5, h(A)=7, h(B)=1

B. h(S)=5, h(A)=5, h(B)=3

C. h(S)=7, h(A)=5, h(B)=1

D. h(S)=6, h(A)=6, h(B)=2

0 **TRUE/FALSE** 2 / 2

The Discriminator network in a GAN is trained until it can no longer distinguish between the fake data produced by the Generator and the real data.

- True
- False